

Venkatesh Vemala - Personal RAG Knowledge Base

Structured profile, professional experience, education, academic record, skills, projects and research context

Prepared from the supplied CV, FAU examination record, and project details provided in this conversation.

PRIVATE RAG DATASET: This document intentionally contains personal contact details and academic identifiers from the supplied documents. Do not publish or share it publicly without redaction.

1. Identity and Contact Information

Full name: Venkatesh Vemala

Professional title: AI Engineer

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Phone: +49 15510526814

LinkedIn: linkedin.com/in/venkatesh-vemala

Current country: Germany

Current city in CV: Erlangen, Germany

Date of birth: 7 May 2000

Place of birth: Nagalapuram, Andhra Pradesh, India

Private postal address shown on FAU record: OG 12, Apt. 113, Am Europakanal 36, 91056 Erlangen

FAU student registration number: 23383701

2. Professional Profile

Profile summary: AI Engineer with hands-on experience in NLP, LLMs and healthcare analytics. Experienced in fine-tuning transformer models such as BioBERT for radiology exam classification, automating data-quality insights, and building analytical dashboards using Qlik and Power BI. Previous experience as a GCP Data Engineering Associate at Accenture provides a foundation in data manipulation, preprocessing and reporting across end-to-end AI and analytics workflows.

3. Professional Experience

3.1 Siemens Healthineers - AI Engineer / Data Analyst

Period: January 2024 - Present

Location: Erlangen, Germany

Focus: AI Engineering (Agentic AI focus) and Data Analytics

- Enhanced BioBERT for precise radiology body-part classification in teamplay insights.

- Developed mismatch-detection logic and derived data-quality insights from outputs; the CV reports 87% accuracy.
- Applied early agentic AI patterns by modularizing validation, consistency checking and insight generation into semi-autonomous processing steps.
- Investigated LLM and NLP techniques for automating exam-description interpretation.
- Documented findings, model behavior and performance metrics for internal teams and stakeholders.
- Built Qlik Sense dashboards to visualize KPIs, mismatch trends and scanner-level performance insights in teamplay.
- Collaborated with stakeholders to translate requirements into analytical dashboards.
- Performed data cleaning, transformation and modeling to support accurate reporting across teams.

3.2 Accenture - Data Engineering Associate

Period: September 2022 - October 2023

Location: Bengaluru, India

- Manipulated and transformed datasets in Google Cloud Platform using BigQuery, Dataflow and Google Cloud Storage based on team requirements.
- Wrote SQL queries and transformation logic for reporting needs.
- Worked with senior data engineers to implement assigned data-processing tasks.
- Validated processed data, resolved minor quality issues and ensured accuracy in delivered datasets.

4. Education

4.1 M.Sc. Data Science - Friedrich-Alexander-Universitaet Erlangen-Nuernberg (FAU)

Study period shown in CV: October 2023 - Present

Location: Erlangen, Germany

Specialization in CV: Artificial Intelligence / Machine Learning

Degree program on transcript: Master of Science: Data Science (20212)

Transcript date: 19 August 2026

Exam status: PV / commenced

ECTS accumulated on transcript: 85 of 120 ECTS

Current average grade on transcript: 2.9 (not final)

Official transcript note: The transcript states that the final grade will not be calculated until all required ECTS credits and course achievements are completed, and that the English transcript is for informational purposes while the German version is legally binding.

4.2 B.E. Computer Science - Bharath University

Study period: August 2018 - May 2022

Location: Chennai, India

Specialization: Software Development and System Design

5. FAU Master's Academic Record - Passed Examinations

The following entries are the detailed examinations explicitly shown as passed in the supplied FAU record. Grades use the German grading scale.

Course	Semester	Date	Grade	ECTS	Attempt
Data-driven methods for dynamical systems	Winter 24/25	10 Feb 2025	4.0	5	1
Knowledge discovery in databases with tutorial	Winter 24/25	17 Feb 2025	3.3	5	2
Pattern recognition	Winter 23/24	9 Apr 2024	2.3	5	1
Pattern analysis	Summer 24	25 Jul 2024	3.7	5	1
Machine learning for time series	Winter 24/25	28 Mar 2025	3.7	5	1
Introduction to control and machine learning	Winter 24/25	10 Feb 2025	2.0	5	1
Decision Theory	Summer 25	24 Jul 2025	3.3	5	1
Mathematical foundations of artificial intelligence, neural networks and data analytics	Winter 25/26	28 Nov 2025	3.3	5	3
Machine learning in signal processing	Summer 25	1 Oct 2025	2.0	5	2
Applied data science in medicine & psychology	Summer 25	21 Jul 2025	1.3	5	1
AI in medical robotics	Winter 24/25	15 Apr 2025	3.0	5	1
Movement Neuroscience: Connections between brain and muscles in humans	Summer 25	8 Aug 2025	4.0	5	2
Human computer interaction	Summer 25	15 Sep 2025	2.7	5	1
Data science survival skills	Summer 24	2 Oct 2024	1.7	5	2
Mathematics of learning	Winter 25/26	24 Feb 2026	4.0	5	3
Master's seminar (Data science)	Winter 24/25	17 Mar 2025	2.0	5	1
Selected topics in mathematics of learning	Winter 25/26	9 Feb 2026	3.7	5	1

6. FAU Specialization / Area Summaries Shown on Transcript

- Specialisation Mathematical Theory / Foundations of Data Science: grade 4.0, 5 ECTS shown at area level.
- Specialisation Data Bases and Knowledge Representation: grade 3.3, 5 ECTS shown at area level.

- Specialisation Machine Learning / Artificial Intelligence: dated 22 July 2026, grade 3.05, status passed, 30 ECTS.
- Specialisation Mathematical Statistical Data Analysis: grade 2.0, 5 ECTS shown at area level.
- Application fields: dated 15 September 2025, grade 2.74, status passed, 20 ECTS.
- Medical Data Science: dated 21 July 2025, grade 1.3, status passed, 5 ECTS.
- Artificial Intelligence in Biomedical Engineering: dated 15 September 2025, grade 3.23, status passed, 15 ECTS.
- Technical key qualifications: dated 2 October 2024, grade 1.7, status passed, 5 ECTS.

7. Modules Listed in the Transcript Without a Completed Passed Result in the Visible Record

The FAU document explicitly notes that not-yet-completely-passed modules may be listed together with passed exams. The following module names appear without a complete passed result in the supplied transcript text:

- Discrete optimization I
- Topics in Time Series Analysis
- Discrete optimization II
- Information theory and coding
- Process-oriented information systems
- Advanced deep learning
- International Information Systems
- Business intelligence
- A look inside the human body - gait analysis and simulation
- Cognitive neuroscience for AI developers
- Robotics in surgery and diagnostics
- Deep learning
- Additional achievements

8. FAU Grading System and Transcript Status Codes

- 1.00 / 1.30 = very good
- 1.70 / 2.00 / 2.30 = good
- 2.70 / 3.00 / 3.30 = satisfactory
- 3.70 / 4.00 = sufficient

Code	Meaning
AN	annulled achievement
AR	accredited achievement
AT	medical certificate
BE / EB	pass
EF	excused absence
PV	commenced
NB	fail
EN	failed at the final attempt
FR	failed to meet the deadline
KO	examination procedure suspended
NE	failed to attend examination / registration cancelled

NT	failed to meet attendance requirements
RT	withdrawal by student
RW	rejection by examiner
TA	fraud
VM	deadline not met with reason
VO	deadline not met without reason for graded achievements
VOU	deadline not met without reason for ungraded achievements
ZNA	certificate not collected

9. Technical Skills

- Machine Learning algorithms
- Generative AI
- Image classification, segmentation and feature extraction
- Google Cloud Platform
- BioBERT and BERT-based models
- NLP
- LangChain
- LLM / RAG data pipelines
- PyTorch
- Agentic AI
- Deep Learning
- Python
- Qlik Sense
- ML pipelines and experimentation
- SQL and query optimization

10. Project Experience

10.1 Brain Tumor Classification System

Technologies / domain: Computer Vision, Deep Learning, Flask, VGG16, TensorFlow/Keras

- Built a CNN-based classification model using VGG16 to detect four MRI classes: Glioma, Meningioma, Pituitary and No Tumor.
- Preprocessed and augmented MRI datasets and trained the model using TensorFlow/Keras.
- Deployed the model as a Flask web application that accepts MRI uploads and returns real-time tumor predictions with confidence scores.

10.2 NexaFlow LLM Fine-Tuning Research Project

Independent research project discussed in this conversation. NexaFlow Technologies is a fictional company created for the experiment.

- Research goal: test whether a small base LLM can be adapted to fictional company policies while reducing unsupported answers and improving policy-aware behavior.
- Base model: Qwen2.5-1.5B base model.
- Training pipeline: synthetic policy corpus -> data engineering -> Continued Pretraining (CPT) + QLoRA -> merge -> Supervised Fine-Tuning (SFT) + fresh QLoRA -> merge -> Direct Preference Optimization (DPO) + fresh QLoRA -> merge -> final benchmark.

- SFT data included known company facts, policy application, numerical rules, unsupported questions, unknown people, false-premise questions, harmful/safety requests, benign security requests and out-of-scope questions.
- Final automatic behavior evaluation: Base 23.3%, CPT 14.7%, SFT 83.3%, DPO 81.3%.
- Main finding: SFT delivered the largest behavioral improvement; DPO produced mixed refinements rather than universally improving all metrics.
- Manual testing also identified limitations in novel multi-step policy reasoning and threshold application.
- Built a Streamlit chatbot prototype for local inference and conversational UI.
- Published project code to GitHub and prepared merged CPT, SFT and DPO models for Hugging Face sharing.

11. Certifications

- Google Cloud Certified - Associate Cloud Engineer
- Understanding Agentic AI / Agentic AI

12. Soft Skills

- Leadership and critical thinking
- Attention to domain details
- Continuous learning
- Teamwork in cross-functional settings

13. Languages

English: C1

German: A1

14. Timeline Summary

- May 2000: Born in Nagalapuram, Andhra Pradesh, India.
- August 2018 - May 2022: B.E. Computer Science at Bharath University, Chennai.
- September 2022 - October 2023: Data Engineering Associate at Accenture, Bengaluru.
- October 2023 - Present: M.Sc. Data Science at FAU, Erlangen.
- January 2024 - Present: AI Engineer / Data Analyst at Siemens Healthineers, Erlangen.
- 19 August 2026: FAU transcript shows 85 of 120 ECTS accumulated and current average grade 2.9 (not final).

15. RAG-Friendly Fact Blocks

The following short fact blocks are intentionally redundant and explicit so that a retrieval system can match common question formulations more reliably.

- Venkatesh Vemala is an AI Engineer based in Germany, with current professional and academic activity in Erlangen.

- Venkatesh has experience in NLP, LLMs, BioBERT, healthcare analytics, Qlik Sense, Google Cloud Platform, Python, PyTorch, SQL, LangChain, RAG pipelines and agentic AI.
- Venkatesh works at Siemens Healthineers as an AI Engineer / Data Analyst. The CV lists the role from January 2024 to the present in Erlangen, Germany.
- Before Siemens Healthineers, Venkatesh worked at Accenture as a Data Engineering Associate from September 2022 to October 2023 in Bengaluru, India.
- Venkatesh is studying an M.Sc. in Data Science at Friedrich-Alexander-Universitaet Erlangen-Nuernberg, with an AI / Machine Learning specialization.
- The FAU transcript dated 19 August 2026 reports 85 ECTS out of 120 and a current average grade of 2.9, which is not final.
- Venkatesh completed a B.E. in Computer Science at Bharath University from August 2018 to May 2022, specializing in Software Development and System Design.
- A key project is a Brain Tumor Classification System using VGG16, TensorFlow/Keras and Flask for MRI-based classification.
- Another key project is the NexaFlow LLM Fine-Tuning research project using Qwen2.5-1.5B with CPT, SFT and DPO stages.
- The strongest automatic evaluation result in the NexaFlow experiment was the SFT model at 83.3% automatic behavior accuracy; DPO scored 81.3%.
- Venkatesh is Google Cloud Certified as an Associate Cloud Engineer.
- Language proficiency listed in the CV: English C1 and German A1.

16. Suggested Questions This RAG Document Should Answer

1. Who is Venkatesh Vemala?
2. What is Venkatesh currently studying?
3. Which university does Venkatesh attend?
4. What subjects has Venkatesh passed in the Data Science master's program?
5. What is Venkatesh's current ECTS total and average grade?
6. What machine learning and AI courses has Venkatesh completed?
7. What professional experience does Venkatesh have?
8. What did Venkatesh do at Siemens Healthineers?
9. What did Venkatesh do at Accenture?
10. Which cloud technologies does Venkatesh know?
11. What projects has Venkatesh completed?
12. What is the NexaFlow fine-tuning project?
13. What are Venkatesh's technical skills, certifications and languages?

17. Source and Data-Quality Notes

- The CV is the source for employment history, profile summary, technical skills, projects, certificates, soft skills and languages.
- The FAU examination record is the source for date/place of birth, private postal address, student registration number, academic status, ECTS, current average grade and detailed examination results.

- The transcript itself states that modules not yet completely passed may appear alongside passed modules; therefore this document separates explicitly passed examinations from module names that appear without a completed passed result.
- The transcript states that the English version is informational and only the German version is legally binding.
- The NexaFlow research-project section is based on project details supplied during this conversation and is not taken from the uploaded CV or transcript.
- For a public or shared RAG system, create a redacted copy that removes the private address, phone number, email and student registration number.